

Blood Pressure Data Collection

PulseDB-aligned ECG and PPG acquisition

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1. What the team collects

Each usable record combines a 10-second ECG/PPG segment with one paired upper-arm cuff reading. The device saves the waveforms. The operator records the cuff values and matching segment ID on paper. The shared ID joins the two records.

Recorded by	Information
Device	Two synchronized channels, 125 samples/second, 1,250 samples per channel per segment.
Paper form	Participant token, visit, cycle, segment ID, SBP, DBP, completion status and relevant acquisition notes.
Session settings	Device models, firmware, channel units, PPG LED channel, sample rate, polarity and timing configuration.

Collection sequence

Use the approved study procedure and cleared body-connected equipment. Assign a random token, confirm the visit number, and prepare the sensors and cuff. After the required seated rest, check both live signals. Save the ten seconds immediately before cuff inflation, then record the displayed SBP and DBP against that segment ID. Keep the participant seated and still during the pair. The plan is three paired cycles per visit and two visits 3-14 days apart. Use the approved interval between cuff measurements. Log an unsuccessful cycle and its reason; a permitted repeat receives a new segment ID. The waveform used for a pair ends before inflation, so cuff-induced disturbance is outside that segment. At the end of the visit, match every paper row to an electronic file, verify the file opens, and transfer the files to approved restricted ND storage. A photograph supports transcription of the paper form; the electronic waveform files remain necessary.

2. File formats and exact fields

The CSV files are project interfaces. PulseDB itself is distributed as MATLAB .mat arrays; its official subset stores ECG, PPG and ABP in Subset.Signals, and labels in Subset.SBP and Subset.DBP. We use ECG and PPG as predictors. ABP supplies the public reference labels.

A. Electronic raw segment

```
time_s,ecg_raw,ppg_raw
```

Exactly 1,250 numeric rows. Time advances by 0.008 seconds. ECG and PPG are measured at the same timestamps; their raw units are documented in the session settings. Retain the continuous source recording.

B. Transcribed paper log

```
participant_id,visit,cycle,segment_id,  
waveform_file,SBP,DBP,status
```

This header is one line in the actual file. waveform_file is the relative file path. SBP and DBP are in mmHg. Use accepted for a complete usable pair. Keep visit/cycle information in this log; it is not a model input.

C. Model-ready wide CSV

Column positions	Exact names / rule	Meaning
1-4	dataset, participant_id, segment_id, fs_hz	Source, person, segment and sampling rate
5-6	SBP, DBP	Reference labels in mmHg
7-1256	ecg_0000 ... ecg_1249	ECG samples in time order
1257-2506	ppg_0000 ... ppg_1249	PPG samples in time order

All 2,506 exact column names are supplied in complete_header.csv. column_dictionary.csv defines every column individually. Sample index j corresponds to $j/125$ seconds. The two signal blocks contain filtered, min-max-normalized amplitudes, without physical units. Labels and identifiers are excluded from the model inputs.

3. Twenty real segment records

The following are the first 20 eligible development-training rows selected by the export script, not simulated data. Each row represents a different 10-second segment from the same public participant. The full CSV also includes all 2,500 waveform values for each row.

Row	Participant	SBP	DBP
1	p000001_1	130.71	87.99
2	p000001_1	129.74	89.15
3	p000001_1	131.32	83.79
4	p000001_1	157.72	104.95
5	p000001_1	123.77	76.92
6	p000001_1	130.37	86.78
7	p000001_1	119.83	80.51
8	p000001_1	126.36	85.51
9	p000001_1	132.65	86.41
10	p000001_1	121.11	76.66
11	p000001_1	133.50	83.99
12	p000001_1	160.56	105.70
13	p000001_1	112.99	74.76
14	p000001_1	148.09	95.13
15	p000001_1	118.82	78.91
16	p000001_1	112.82	66.31
17	p000001_1	110.42	74.19
18	p000001_1	119.83	82.93
19	p000001_1	116.07	69.26
20	p000001_1	117.57	77.25

Read the first row: participant p000001_1, segment official_train_row_1. Its labels are 130.71/87.99 mmHg. The model receives that row's ECG/PPG arrays and predicts two blood pressure values. Files: real_20_segments.csv contains the complete rows; real_20_segments_metadata.csv is the compact view above. Row numbers in segment IDs refer to the official MATLAB training subset (one-based indexing), not the original source-record SegmentID. provenance.json records the extraction details.

4. Reading the waveform values

This is a second view of the first segment above. Here one row is one sampling instant, not one segment. The first 20 points cover 0.000-0.152 seconds.

Index	Time (s)	ECG	PPG
0	0.000	0.127385	0.038952
1	0.008	0.144529	0.049507
2	0.016	0.161971	0.064195
3	0.024	0.183065	0.083477
4	0.032	0.201643	0.107861
5	0.040	0.210779	0.137500
6	0.048	0.212760	0.172536
7	0.056	0.204401	0.212786
8	0.064	0.176399	0.257863
9	0.072	0.135182	0.307352
10	0.080	0.099445	0.360438
11	0.088	0.078871	0.416186
12	0.096	0.074263	0.473611
13	0.104	0.082308	0.531529
14	0.112	0.093708	0.588727
15	0.120	0.101428	0.644077
16	0.128	0.101906	0.696453
17	0.136	0.094458	0.744762
18	0.144	0.110324	0.788136
19	0.152	0.218902	0.825858

For example, sample 1 is stored in `ecg_0001` and `ppg_0001` in the wide file. Both correspond to 0.008 seconds. Connecting successive values against time produces the plotted waveform. These are normalized signal amplitudes, not blood pressure values. The actual exported values retain float32 precision. This page rounds to six decimal places for readability. `real_20_samples.csv` contains the full numeric values and the matching participant/segment IDs.

5. Paper collection record

Print one sheet per visit. Use a random token, not a name. The participant retains the same token for the return visit.

Participant token: _____ Visit: _____

Acquisition configuration ID: _____

Cycle	Segment ID	SBP	DBP	Status / note
1				
2				
3				
Repeat				
Repeat				
Repeat				

Before leaving the session

Confirm that each complete row has a matching waveform file and legible cuff values. Mark incomplete or repeated cycles and keep their notes. Retain the original record when correcting a transcription. For photo-based transcription, photograph the full sheet clearly and verify the returned numeric table against it. Example naming: ND017_V1_C2 is participant ND017, visit 1, cycle 2. Save its electronic segment as ND017_V1_C2.csv. A repeat can use ND017_V1_C2_R1. The operator records the exact same ID on paper. Actual participant files and photographs belong in restricted study storage. The public GitHub nd folder holds templates and conversion instructions only.

6. Conversion and prediction

From the model directory, install requirements in a Python environment. The trained selection is `cnn_full`, a two-channel residual CNN.

```
python predict.py "../../data/PulseDB/examples/real_20_segments.csv"
    --output predictions.csv --format filtered
python convert_nd.py /restricted/reference_log.csv
    --output /restricted/nd_segments.csv
python predict.py /restricted/nd_segments.csv
    --output /restricted/nd_predictions.csv --format filtered
```

Commands are wrapped here; enter each command on one line. On macOS, use `sh run.sh predict.py` ... if XGBoost/OpenMP loading requires the included launcher. The selected CNN itself does not need XGBoost. The ND adapter filters ECG at 0.5-40 Hz (fourth-order Butterworth) and PPG at 0.5-8 Hz (fourth-order Chebyshev II, 20 dB stopband parameter), then normalizes each channel to 0-1. The source uses provided filtered channels; this window-local adapter is a defined deployment implementation, not a claim of exact equivalence to the source filter chain.

Measured model performance

Evaluation	SBP MAE	DBP MAE
Official unseen-person test	12.46	8.46
Prior-exposure-excluded test	12.44	8.39
AAMI-designated subset	19.34	13.17

All errors are mmHg. The official test has 144 people/57,600 segments. Three people appeared in earlier exploratory work; excluding them leaves 141 people/56,400 segments. The AAMI-designated subset has 116 people/666 segments; its name is not a certification. The model is a research predictor, with ND transfer performance still to be measured. Raspberry Pi latency has not been measured. Sources: [PulseDB repository](#); [official author-hosted subsets](#); Wang et al., *Frontiers in Digital Health* (2023), DOI: 10.3389/fdgth.2022.1090854. Public sample data: CC BY-NC-SA 4.0; license and attribution accompany the examples.

7. Complete V1 error metrics

V1: high-resolution two-channel residual 1D CNN, trained with standardized-target MSE. These metrics use saved predictions from the selected checkpoint; the weights are unchanged.

Official calibration-free test: 144 people / 57,600 segments

Metric	SBP	DBP
MAE	12.46	8.46
Bias	-1.92	-1.64
Error SD	15.89	10.66
Within 5 mmHg (%)	26.06	37.24
Within 10 mmHg (%)	48.80	66.11
Within 15 mmHg (%)	67.55	84.62
Participant-macro MAE	12.46	8.46
Participant MAE: median	11.02	7.32
Participant MAE: 90th percentile	20.80	14.35

AAMI-designated subset: 116 people / 666 segments

Metric	SBP	DBP
MAE	19.34	13.17
Bias	-12.74	-9.31
Error SD	22.72	13.93
Within 5 mmHg (%)	19.22	24.77
Within 10 mmHg (%)	36.19	45.20
Within 15 mmHg (%)	53.00	67.27
Participant-macro MAE	19.58	13.10
Participant MAE: median	15.95	11.46
Participant MAE: 90th percentile	36.76	23.72

Error is prediction minus reference; SD uses N-1. Errors are in mmHg except percentages. Thresholds are inclusive. Participant-macro MAE weights each person equally. These are source-data evaluations, not a clinical validation of a cuff device. Full validation and exposure-excluded metrics, plus each participant's MAE, are in the V1 reports folder.